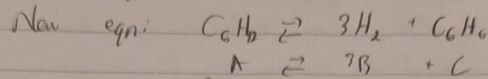


P6.1:

c) Example 6.2:



New $K_c = 0.001 \left(\frac{\text{mol}}{\text{L}} \right)^3 \quad (k_{eq})$

Due to new stoichiometry, $\frac{r_A}{-1} = \frac{r_B}{3} = \frac{r_C}{1}$

$\Rightarrow \frac{dF_A}{dV} = r_A$

$\frac{dF_B}{dV} = -3r_A - k_c T_{T_0} \frac{F_B}{F_T}$

$\frac{dF_C}{dV} = -r_A$

$K_c = \frac{C_B^3 C_C}{C_A} \Rightarrow C_A = \frac{C_B^3 C_C}{K_c} = \frac{C_{T_0}^4 \left(\frac{F_B}{F_T} \right)^3 \frac{F_C}{F_T}}{K_c}$

$= \frac{C_{T_0}^4}{K_c} \frac{F_B^3 F_C}{F_T^4}$

$r_A = -k_c C_{T_0}^4 \left[\frac{F_A}{F_T} - \frac{C_{T_0}^3}{K_c} \frac{F_B^3 F_C}{F_T^4} \right]$

$\rightarrow$ From graph:

- higher methane efflux rate, R_B throughput reactor
- higher flow rate F_B throughput reactor
- slightly lower convergen throughput, lower convergence

at the end

d) Ex 6-3:

1) Varying C_{B_0} :

- reach 100% conversion earlier as C_{B_0} increase
- r_A amplitude increases as C_{B_0} increases
- $C_{B_0} = 0 \rightarrow$ no conversion & $r_A = 0$ throughout reactor

Varying V_0 :

- increasing $V_0 \Rightarrow$ decrease conversion
- V_0 (dm³ min⁻¹): conversion reached very quickly, r_{Amax} has

highest peak.

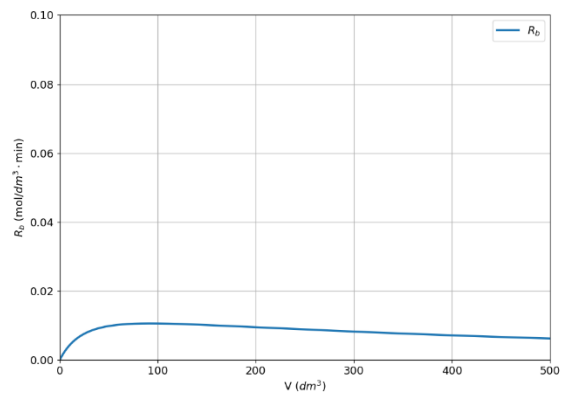
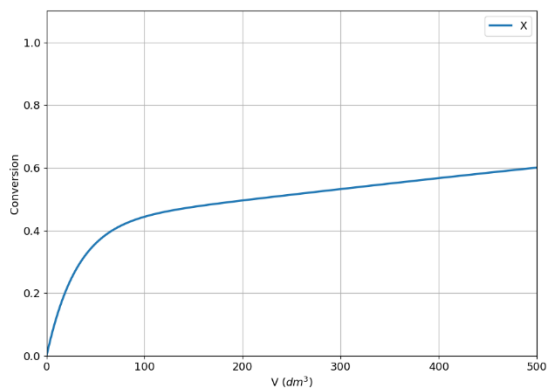
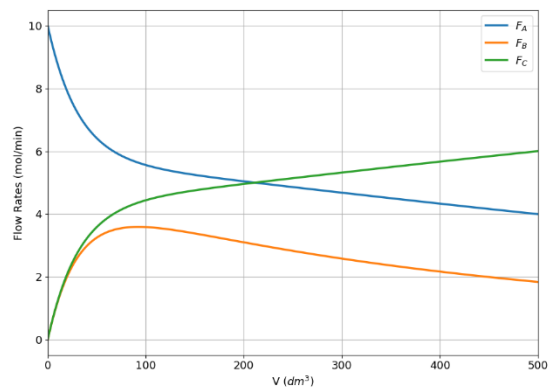
$V_0 = 40$ dm³ (max) $\Rightarrow$ lower overall conversion, $\sim 30\%$, r_{Amax} has much lower value, C_A decrease less over time.

Varying U_0 :

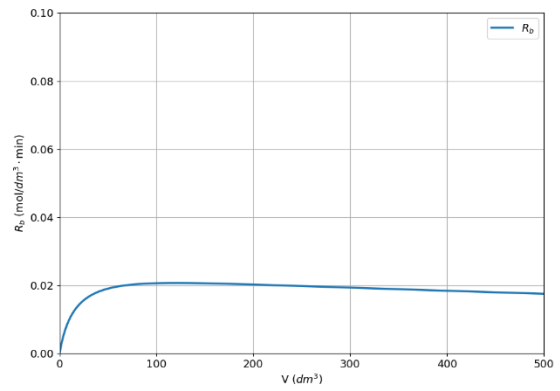
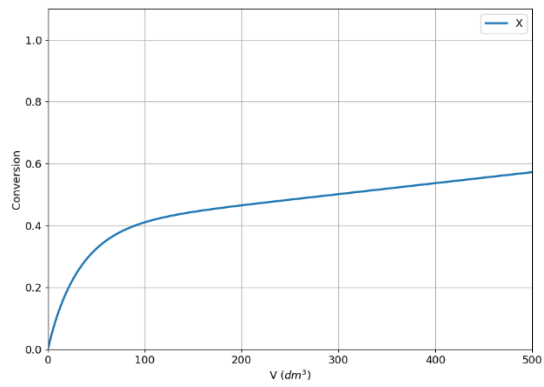
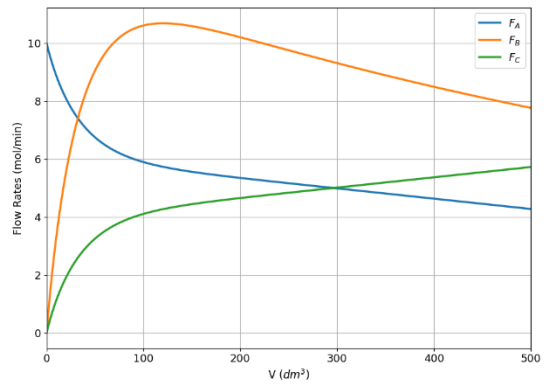
- increasing U_0 makes conversion reach 100% earlier
- higher peak value for r_A

2) Conclusion:

- increasing C_{B_0} will make conversion reach 100%, if it can reach already, increasing further would make time reaching 100% conversion shorter
- increasing C_{B_0} also increase reaction rate r_{Amax}
- Increasing V_0 decrease conversion and r_{Amax}
- Increasing k has similar effect as with C_{B_0} , qualitatively.
- Increasing U_0 also yield similar effect with increasing C_{B_0} qualitatively



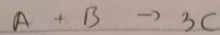
Graphs for example 6-2



Graph for new stoichiometry and reaction rate

e 2) Example 6-4:

C_{A_0} & C_{B_0} have equally greatest effect on convergence



$\therefore$ makes sense

2) C_{A_0} & C_{B_0} variation can cause pressure drop to atmosphere at the outlet due to change in both would affect F_{T_0} and F_T

This can be done by keeping either C_{A_0} or C_{B_0} constant and let the other to 0.045 mol/L (keep α and k as given values)

We can also increase α and/or k then adjust other parameter accordingly.

3) Conclusions

- Varying C_{A_0} & C_{B_0} while keeping the other constant gives the same effect
- increasing convergence with lower pressure drop ratio
- increasing convergence at higher pressure drop will increase equilibrium constant
- increasing convergence and higher pressure drop with increasing C_{A_0} or C_{B_0}
- varying C_{A_0} and C_{B_0} have same effect on concentration profile, convergence and pressure drop due to reaction stoichiometry